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Task 3.3

The class `HashTable` contains 11 buckets and uses linear probing (seeks the next available slot in the hash table by probing sequentially, skipping **2 buckets at a time**) for its collision resolution. It has two attributes:

- `size`, the number of buckets in the Hash Table.
- `array`, a 1-dimensional list that is used to store either `Student` or `Staff` objects. Each element in the list is initialised to `-1` in the beginning.

The class `HashTable` has the following methods:

- a constructor that initialises `size` and `array` to appropriate values for an empty Hash Table. It also initialises the number `-1` for each element in `array`.
- `hash(nric)` takes the `nric` (string) as a parameter and returns the position in the `array` to store the data (`Student` or `Staff` Object). The hash algorithm is shown in the pseudocode below:

```
h ← 0
FOR i ← 0 TO length(nric) - 1
    val ← 57 * (ASCII value of nric[i])
    h ← (h + val) % size
NEXT i
RETURN h
```

- `insert(Object)` takes a `Staff` or `Student` object as a parameter and inserts it into the `array` based on the calculated position returned by the above hash algorithm. If there is a collision, use linear probing to seek the next available slot (Note: to skip 2 buckets at a time). Return `False` if unable to find any available bucket, otherwise return `True`.
- `displayInformation(nric)` takes the `nric` (string) as a parameter and calls `display()` method of the `Staff` or `Student` Object that contains that `nric` value. Print "Person not found" if unable to locate any `Staff` or `Student` object with that `nric` value.

Write program code to declare the class `HashTable` with all its attributes and methods.

Task 3.4

The csv file, `SchoolInfo.csv`, contains information of a number of Staff and Students. Each row is a comma-separated list of data of the following:

- Text identifier (indicating whether the row belongs to a Student or Staff)
- NRIC
- Full Name
- Date of Birth in the form YYYY-MM-DD
- Class (for Students), Role (for Staff)

Write a program to:

- Create an empty `HashTable` instance.
- Read in the information from the csv file, creating an instance of the `Student` class or `Staff` class based on the Text identifier.
- Insert all `Student` and `Staff` instances into the `HashTable` instance. Output "Insert successful" if successfully inserted into the `HashTable`. Otherwise, output "Insert unsuccessful".
- Prompt user for an input to key in an `nric` value and call the `displayInformation()` method of the `HashTable` instance to display the

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information of the `Student` or `Staff` with that `nric`.

- Ask the user if they would like to search again. Repeat the process of prompting user to key in an `nric` value and displaying information of the `Student` or `Staff` instance if User input "Yes".

Test your program by running the application with a `Student nric`, a `Staff nric` and an `nric` value that is not found in `SchoolInfo.csv`. [5]

Save your Jupyter Notebook for Task 3.